SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

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QUESTIONS 3 / 10 completed

Double Ended Queue

DOUBLE ENDED QUEUE

Write a C program to implement a Double Ended Queue (Deque) using Arrays where insertion and deletion can be performed from both front and rear ends. Consider a warehouse inventory management system where goods can be added or removed from either side based on priority. Sample Input: Insert Front 100, Insert Rear 200, Insert Front 50, Delete Rear. Sample Output: Deleted Element: 200. Queue Elements: 50 100.

PROGRAM EDITOR

C ✓ Updated Run Save

```c
#include <stdio.h>
#define MAX 100

int deque[MAX];
int front = -1, rear = -1;

void insertFront(int value) {
    if (front == -1) {
        front = rear = 0;
        deque[front] = value;
    } else if (front == 0) {
        printf("Overflow: Cannot insert at front\n");
        return;
    } else {
        front--;
        deque[front] = value;
    }
}

void insertRear(int value) {
    if (front == -1) {
```

OUTPUT

```
Overflow: Cannot insert at front
Deleted Element: 200
Queue Elements: 100
```

INPUT

Insert Front 100
Insert Rear 200
Insert Front 50